

Dirac Notation and State Vectors

22, 24 September · Objective 3 · Quiz 1 this week

Dirac notation is an interface specification. It separates *what a state is* from *how it is represented*, so that operations can be written once and evaluated in whatever basis is convenient. Treat it as typed notation and the algebra becomes mechanical.

Kets, bras, and types

A **ket** $|\psi\rangle$ is a vector in a complex Hilbert space — for one qubit, a column in $\mathbb{C}^2$:

$$\begin{array}{lll} |\mathbf{0}\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} & |\mathbf{1}\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} & |\psi\rangle = \alpha|\mathbf{0}\rangle + \beta|\mathbf{1}\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{array}$$

A **bra** $\langle\phi|$ is the dual — the conjugate transpose, a row vector:

$$\langle\psi| = |\psi\rangle^\dagger = [\psi_0^* \quad \psi_1^*]$$

The value of the notation is that the type of every expression is readable from its shape:

EXPRESSION	SHAPE	TYPE	MEANING
$\langle\psi \psi\rangle$	row $\times$ column	scalar ($\mathbb{C}$)	overlap / amplitude
$ \psi\rangle\langle\psi $	column $\times$ row	operator (2 $\times$ 2)	outer product
$\hat{U} \psi\rangle$	matrix $\times$ column	ket	evolved state
$\langle\psi \hat{A} \psi\rangle$	row $\times$ matrix $\times$ column	scalar ($\mathbb{R}$ if $\hat{A}$ Hermitian)	expectation value

Analogy boundary — bra/ket as a typed contract. Kets are the state type, bras the measurement-side consumer, and $\langle\phi|\psi\rangle$ the only well-typed pairing that returns a scalar. The analogy is genuinely load-bearing: it prevents the most common algebra errors, because ill-typed expressions are visibly ill-shaped. Where it breaks: your interfaces are usually covariant in one direction, while the bra–ket relationship is **antilinear** — $\langle\phi|$ for the ket $c|\phi\rangle$ is $c^*\langle\phi|$, with the conjugate. Lose the conjugate and probabilities stop being real.

The inner product and the Born rule

$$\langle\psi|\psi\rangle = \psi_0^*\psi_0 + \psi_1^*\psi_1 \in \mathbb{C}$$

Properties: $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$, linear in the ket, antilinear in the bra, and $\langle\psi|\psi\rangle = 1$ for normalized states.

The Born rule in this notation is a single line:

$$P(\text{outcome } \psi) = |\langle\psi|\psi\rangle|^2$$

Orthogonal states ($\langle\phi|\psi\rangle = 0$) are perfectly distinguishable; identical states have overlap 1. Everything between is the degree to which one measurement can confuse two states — which is exactly what no-cloning turned on last week.

Operators, Hermiticity, and unitarity

Two operator classes do all the work:

Hermitian ($\hat{A}^\dagger = \hat{A}$) represent **observables**. Real eigenvalues (measurement outcomes) and orthogonal eigenvectors (distinguishable results). The Pauli operators:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

All Hermitian, all with eigenvalues ± 1 . Z's eigenvectors are $|0\rangle, |1\rangle$; X's are $|\pm\rangle$.

Unitary ($\hat{U}^\dagger\hat{U} = I$) represent **evolution**. Norm-preserving, invertible, and — as established in Week 5 — the only operations quantum mechanics permits between measurements.

The Paulis are both Hermitian *and* unitary (since $X^2 = Y^2 = Z^2 = I$), which is why they serve as observables and as gates. That coincidence is specific to involutory operators; the Hadamard shares it, most gates do not.

The **spectral decomposition** connects the two roles:

$$\hat{A} = \sum_m a_m |m\rangle\langle m|$$

An observable is its eigenvalues weighted by projectors onto its eigenstates. Measurement is the application of one such projector, selected stochastically by the Born rule. $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ is worth verifying by hand.

Expectation values

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle = \sum_m a_m P(m)$$

The mean outcome over many measurements on identically prepared states. Note that this is never what a single run gives you — quantum hardware returns samples, and every observable in Weeks 9–15 is estimated from repeated **shots**. Budgeting shots is a real experimental cost: statistical error falls as $1/\sqrt{N}$.

For $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$: $\langle Z \rangle = 0$ (equal 0s and 1s), $\langle X \rangle = 1$ (certain outcome in the X basis). Those two numbers together locate the state, which is the Bloch-sphere construction of Week 7.

Composite systems

Two qubits combine by tensor product:

$$|\psi\rangle \otimes |\varphi\rangle = |\psi\varphi\rangle, \quad \text{written } |00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Dimension multiplies: n qubits span 2^n dimensions. Operators compose the same way — $X \otimes I$ applies X to qubit 0 and leaves qubit 1 alone.

Analogy boundary — tensor product as a struct. Combining subsystems into a composite resembles a product type, and for *separable* states the analogy holds exactly. It fails for entangled states, which is the whole point: a struct is always the pair of its fields, and you can project out either one losslessly. An entangled state has **no** well-defined field values — the partial trace yields a mixed state, not a pure component. Product types describe only the separable corner of the space, which is a vanishing fraction of it.

Quiz 1 review

Covering history of physics and the physical foundations. The reliable failure points:

- **Superposition is basis-relative; entanglement is not.** Expect a question that hinges on this.
- **Uncertainty is a property of states, not instruments.** Non-commutation, not clumsiness.
- **Which-path information destroys interference regardless of mechanism.**
- **Decoherence explains loss of interference, not outcome selection.**
- **Bell violation refutes local realism jointly** — you may keep either alone.
- **Photoelectric effect: frequency sets energy, intensity sets count.**

Key takeaways

- Dirac notation types every expression by shape: $\langle\phi|\psi\rangle$ is a scalar, $|\psi\rangle\langle\phi|$ an operator.
- The inner product is antilinear in the bra; the conjugate is what keeps probabilities real.
- Hermitian operators are observables (real eigenvalues, orthogonal eigenvectors); unitary operators are evolution (norm-preserving, invertible).
- Spectral decomposition $\hat{A} = \sum a_m |m\rangle\langle m|$ unifies both roles; measurement applies one projector.
- Composite systems combine by tensor product with dimension 2^n ; entangled states have no well-defined component states.

Self-check

Q1 (Objective 3) — Show that $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ by computing the outer products, then explain what goes wrong physically if you write $Z = |0\rangle\langle 0| + |1\rangle\langle 1|$ instead.

Q2 (Objective 3) — Explain why $\langle\phi|\psi\rangle$ must be antilinear in the bra rather than linear in both arguments, using the requirement that probabilities be real and non-negative.

Check your answers

A1 — Compute the outer products directly:

$$\begin{aligned} |0\rangle\langle 0| &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} & |1\rangle\langle 1| &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ |0\rangle\langle 0| + |1\rangle\langle 1| &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \checkmark \end{aligned}$$

Writing $+$ instead gives $|0\rangle\langle 0| + |1\rangle\langle 1| = I$, the identity — the **completeness relation**, not Z .

Physically, the two differ in their eigenvalues, which are the measurement outcomes. Z has eigenvalues $+1$ and -1 , assigning distinct labels to $|0\rangle$ and $|1\rangle$ so the operator distinguishes them; $\langle Z \rangle$ then carries information about the state. The identity has eigenvalue $+1$ for every state — it is the observable that always returns 1 and therefore tells you nothing. $\langle\psi|I|\psi\rangle = 1$ for all normalized ψ .

The structural point: in $\hat{A} = \sum a_m |m\rangle\langle m|$, the projectors specify *which outcomes are distinguishable* and the coefficients specify *what value each outcome is labeled with*. Set all coefficients equal and you have destroyed the observable's ability to discriminate, while keeping a perfectly valid — and useless — Hermitian operator.

A2 — The requirement is that $P = |\langle\phi|\psi\rangle|^2$ be real, non-negative, and **invariant under rephasing**, since global phase is unphysical.

Suppose the inner product were linear in both arguments. Take the ket $c|\psi\rangle$ with $c = e^{i\theta}$. Linearity in the bra would give $\langle c\phi|\psi\rangle = c\langle\phi|\psi\rangle$, and the norm $\langle\psi|\psi\rangle$ for the state $c|\psi\rangle$ would become $c^2\langle\psi|\psi\rangle = e^{2i\theta}$ — a complex number. A state's norm would depend on an unobservable phase, and $|\langle\phi|\psi\rangle|^2$ would no longer be phase-invariant.

Antilinearity fixes this exactly. With $\langle c\phi| = c^*\langle\phi|$:

$$\langle c\psi|c\psi\rangle = c^*c\langle\psi|\psi\rangle = |c|^2\langle\psi|\psi\rangle = \langle\psi|\psi\rangle \quad \checkmark \text{ real, positive, phase-independent}$$

The general statement: an inner product on a complex vector space must be **conjugate-symmetric** ($\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$) for $\langle\psi|\psi\rangle$ to be real, and positive-definite for it to be a probability. Conjugate symmetry plus linearity in one argument forces antilinearity in the other. There is no choice available — it is the

only structure on $\mathbb{C}^n$ supporting a probabilistic reading, which is why the same conjugation appears in the dagger of every operator.